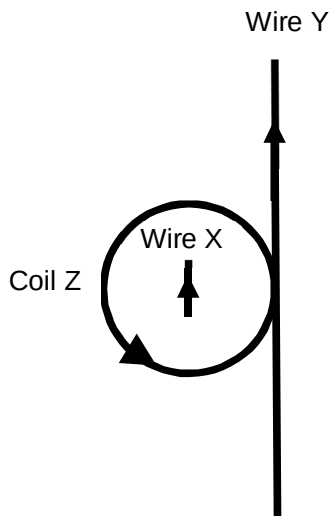


- 24 The diagram below shows a setup of two current-carrying wires X and Y, and a single-turn coil Z.



Wire X has length 0.050 m with a current of 1.0 A flowing through it. Wire Y has length 2.0 m with a current of 0.50 A. Coil Z has a diameter of 0.20 m with a current of 1.5 A.

What is the resultant force on wire X if the centre of coil Z is on wire X?

- A $5.0 \times 10^{-8} \text{ N}$
- B $4.2 \times 10^{-7} \text{ N}$
- C $5.2 \times 10^{-7} \text{ N}$
- D $9.4 \times 10^{-6} \text{ N}$